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Drone People Count

Data Challenges

01 • Small People

A person is ~10–30 px in a 12 MP frame. Downscaling to 640 px erases them entirely.
Solution: SAHI

02 • Clusters & Occlusion

Seated groups overlap, leading to merged bounding boxes and system under-counting.
Solution: tile-merge

03 • RGB vs Thermal Gap

Thermal is WhiteHot, contrast-processed with auto-gain. Standard COCO-trained detectors have never seen this domain.

04 • Sensor Alignment Offsets

Resolution, field-of-view (FOV), and mechanical timing offsets between the two sensors mean image pairs are not pixel-aligned.

05 • Heat Sources & Look-Alikes

Warm non-human objects and person-shaped clutter (bollards, umbrellas) trigger false positives.

Model Research & Selection

YOLO11 + SAHI — One-Stage CNN

Architecture: CNN backbone → PAN feature-pyramid neck → decoupled, anchor-free head.

How it works: One forward pass predicts all boxes densely at multiple scales → NMS de-duplicates. SAHI slices frame into overlapping tiles, detects per tile, and merges back → small people become "large" within a tile.

RT-DETR / RF-DETR — Detection Transformer

Architecture: CNN backbone → transformer encoder–decoder → N learned object queries, each emitting one box.

How it works: Global attention over the whole image; a fixed set of queries predict objects as a set (bipartite matching) → anchor-free and NMS-free, end-to-end.

Decision & Trade-offs

Chosen: YOLO11 + SAHI — Mature pretrained person weights, SAHI tiling for small aerial objects, CPU-viable, first-class fine-tuning API, tunable NMS.

Trade-off: RT-DETR's Apache-2.0 licensing is cleaner; pipeline is written detector-agnostic so swapping is cheap.

Criterion	Model 1 — YOLO11 + SAHI	Model 2 — RT-DETR/RF-DETR
Small aerial objects (tiled)	✓✓	✓✓
RGB + thermal (via preproc)	✓	✓
Pretrained weights	✓✓	✓
Speed / HW cost	✓ (N× tiles, CPU-viable)	~ (heavier, GPU)
Fine-tuning ease	✓✓	~
Licensing	AGPL (+MIT)	**Apache**
Cloud / edge	✓✓	✓

Pipeline Architecture

01 • Ingestion & Prep

The initial orchestration stage managed by `main.py` to ingest and format sensor data.

- **1. Load**
  Read source frame assets
- **2. Modality**
  Identify sensor type (RGB/Thermal)
- **3. Preprocess**
  Apply CLAHE & normalization

02 • Core Detection

Slicing, inference, and filtering algorithms to isolate people in challenging frames.

- **4. Detect**
  YOLO11 + SAHI sliced windowing
- **5. Keep People**
  Filter out non-human categories
- **6. Confidence / NMS**
  De-duplicate overlapping predictions

03 • Output & Save

Final processing stages delivering annotated visuals and structured JSON telemetry.

- **7. Count & 8. Annotate**
  Tally detections & draw visual overlays
- **9. Save JSON**
  Serialize data structures to disk

System Output Assets

- Annotated Image
- Structured JSON Metadata:

`{image_name, modality, people_count, detections[...]}`

Key Parameters & Preprocessing

01 • RGB Modality

Pipeline settings for standard visual spectrum frames.

Processing Mode

Near pass-through pipeline. The SAHI resolution enhancement handles the primary detail preservation.

02 • Thermal Modality

Pre-processing adjustments for infrared sensor inputs.

Transforms

```
grayscale
↓
CLAHE
↓
3-channel
```

Applied as the baseline default configuration.

03 • SAHI Execution

Slicing parameters and detection aggregation logic.

Tiling Config

640-px tiles with 0.2 overlap.

Post-Processing Flow

```
keep person → confidence (RGB 0.25 / thermal 0.20) → NMS IoU 0.5 → count
```

Ground Truth & Validation Method

01 • Point Annotations

One point per person on **8 images** (4 RGB + 4 thermal), hand-made with a **custom annotation tool** (zoom/pan, keyboard-driven).

02 • Why Points, Not Boxes

Counting needs *presence*, not exact extent. Points are faster and unambiguous on tiny or heavily occluded people.

03 • Matching Rule & Metrics

Detection is a **TP** if its box contains an unmatched GT point (greedy by confidence, one-to-one) → precision / recall / F1.

`|pred − GT| → MAE`

Results: RGB is strong

0.885

F1 SCORE

RGB performance is robust with **MAE 5.5** and **precision 0.93**.

Small error even on the densest frames (111 → 100)

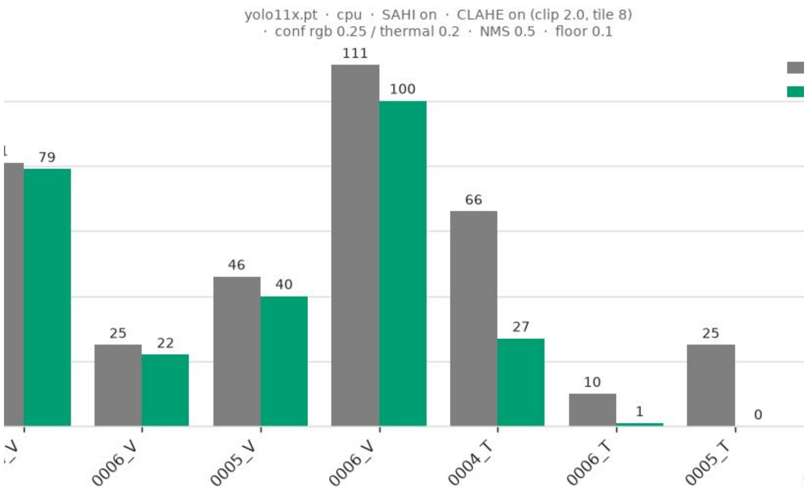
ERROR

ANALYSIS

Errors are primarily **dense-cluster** members and the smallest or occluded people.

Under-counting, not false alarms.

People count: ground truth vs predicted (per image)




RGB GT 111 pred 100 TP 93 FP 7 FN 18

Results: Thermal is the Hard Case

0.25
RECALL (F1: 0.385)

The gap is almost entirely **recall**, not precision.

The model ***misses*** people rather than inventing them.

28
MEAN ABS. ERROR

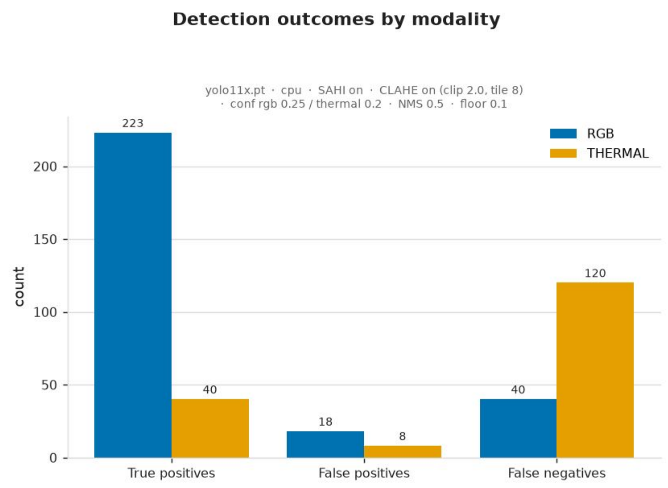
High rate of miscounting.

One specific frame (`0005_T`) resulted in **zero detections**.

COCO
DOMAIN MISMATCH

The model was pre-trained on standard COCO datasets.

Direct application to **`WhiteHot`** thermal causes a severe performance drop.




THERMAL GT 25 pred 0 TP 0 FP 0 FN 25

The ablation: what actually moves thermal

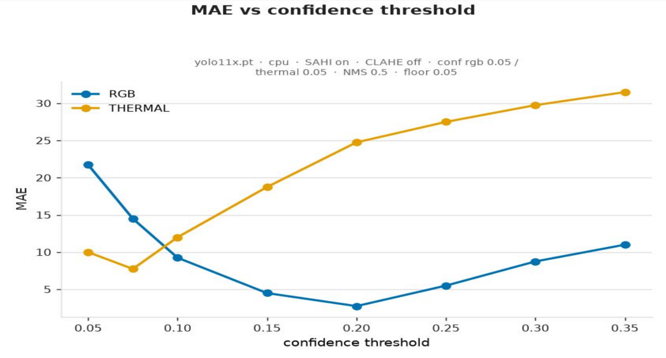
CLAHE Swept full clip × tile grid — **no setting beats "off."** CLAHE mildly hurts thermal. (Kept on as baseline; documented the better setting.)

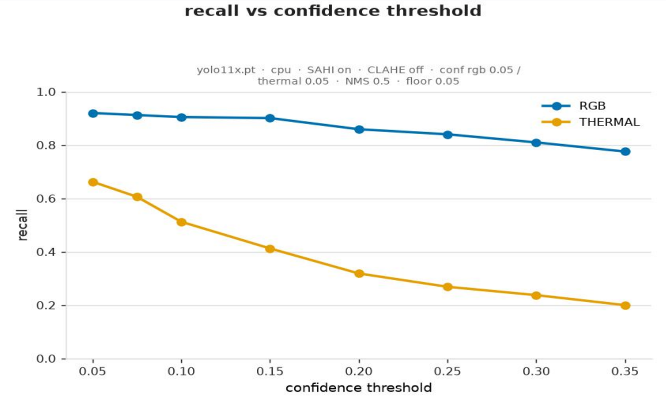
CONFIDENCE **The big lever.** Thermal is under-confident, not blind — fires on ~66% of people. Dropping to ~0.075 **cuts error (MAE 28 → 7.25)** and lifts F1 to 0.673, recall to 0.63.

RGB MODE **Wants the opposite** (higher threshold, ~0.20–0.25). Two optima on opposite ends validate per-modality thresholds.

PRECISION **The ceiling.** Recovering recall floods in false positives → thermal F1 caps ~0.68. Tuning makes thermal usable, not good.

TRUST NOTE **Sweep matches real run** at same threshold (7.75 sweep vs 7.25 real at 0.075) because SAHI's tile-merge was pinned.



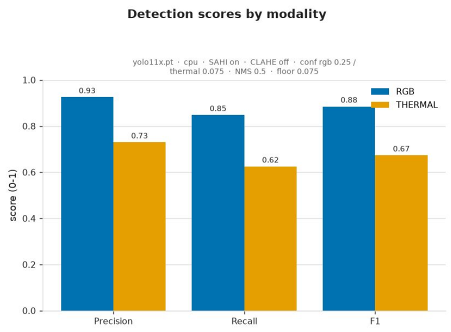


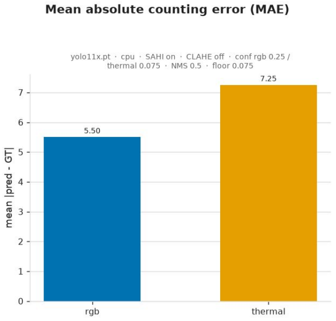
RGB vs thermal comparison

BEST-VS-BEST **RGB still wins on every pair.** RGB (0.25) F1 0.885 / MAE 5.5 vs thermal (CLAHE off, 0.075) F1 0.673 / MAE 7.25 — even with thermal tuned.

WHEN WOULD THERMAL WIN? **Darkness or zero visible light.** Not this dusk scene, and only with a thermal-appropriate detector.

FUSION PROPOSAL (NOT BUILT) rigid mount → fixed homography → use strong RGB to **confirm/seed** thermal detections, recovering thermal FNs.





Limitations

SMALL SAMPLE **Small, single-condition sample.** — 8 images, one location, one dusk session → results are indicative, not generalisable.

DOMAIN MISMATCH **Thermal domain mismatch is the dominant limitation.** Threshold tuning serves as a temporary mitigation, not a permanent fix.

NO FINE-TUNING **A deliberate, evidence-backed choice.** Selecting points instead of bounding boxes means 8 images would easily overfit.

Recommended next steps

01

Fine-tune a thermal detector

FLIR ADAS / aerial thermal

Lifts the whole precision–recall curve in ways that threshold tuning alone cannot.

02

Implement RGB↔thermal fusion

Fixed homography

Establish multi-modal mapping via cross-validation and False Negative (FN) recovery to leverage strong RGB signals.